

Lecture Notes on

Unit 02 Differential Equations

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Within the contents of this is my study guide/ personal notes to MAP2302 which was taken in the Spring of 2026 at Palm Beach State (PBS) Community College in Lake Worth, Florida under Professor Tamara Johns. This examination will be taken on Friday, March 6th 2026 at blah blah blah from the hours of 3pm - 9pm.

This chapter predominantly covers the following topics (General):

1. Lesson 7 Higher-Order Linear Differential Equations
2. Lesson 8 Higher-Order Differential Equations / Reduction of Order
3. Lesson 9 Homogenous Linear Equations with Constant Coefficients
4. Lesson 10 Undetermined Coefficients - Superposition Approach
5. Lesson 11 Undetermined Coefficients - Annihilator Approach

The main revisions made in this study guide versus Unit 01 will be the organization and the utilization of some sort of Appendix to be able to keep track of misc things/ concepts (int. by parts shortcuts to save time, intermediate trig, etc.).

I also want to make use of subsubsections in this paper, so hopefully that goes well. I think that the organizational structure of a toc that utilizes subsubsections looks really clean, as often done by professor David Tong.

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1 Higher Order Linear Differential Equations

1.1 Higher order linear ODEs (Introduction)

1.2 Higher Order Linear ODEs (Continued)

1.3 Second-Order Differential Equations (Introduction)

1.4 How to Compute a Wronskian & Linear Independence

2 Higher Order Linear Differential Equations/ Reduction Of Order

2.1 Reduction of Order (Introduction)

2.2 Reduction of Order (Additional Homogeneous Equation Example)

2.3 Reduction of Order (Second-Order Non-Homogeneous Example)

3 Homogeneous Linear Equations with Constant Coefficients

3.1 Second-Order Homogeneous Equations (Constant Coefficients Introduction)

3.2 Second-Order Equations with Constant Coefficients (Distinct Real Roots)

3.3 Second-Order Equations with Constant Coefficients (Repeated Real Roots)

3.4 Second-Order Equations with Constant Coefficients (Complex Roots)

4 Undetermined Coefficients - Superposition Approach

4.1 Higher order linear ODEs

4.2 Nonhomogeneous equations

4.3 Method of Undetermined Coefficients (Second Order Non-Homogeneous Equations)

4.4 Undetermined Coefficients (Multiplying by x for Linear Independence)

5 Undetermined Coefficients - Annihilator Approach

5.1 How to use the Annihilator Method to Solve a Differential Equation

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